Lagrange equation:

$$T_{pendulum} = \frac{1}{2} m L^2 \dot{\theta}^2$$

- Frame:

$$r = w_1 + L\sin\theta \rightarrow v_{rotation} = \Omega \cdot r = \Omega(w_1 + L\sin\theta)$$

$$T_{frame} = \frac{1}{2} m v_{rot}^2 = \frac{1}{2} m \Omega^2 (w_1 + L\sin\theta)^2$$

$$T = T_{pendulum} + T_{frame} = \frac{1}{2} m L^2 \dot{\theta}^2 + \frac{1}{2} m \Omega^2 (w + L\sin\theta)^2$$

$$V = mg(h - L\cos\theta)$$

$$L = T - V = \frac{1}{2} m \left[l^2 \dot{\theta}^2 + \Omega^2 (w + L\sin\theta)^2 \right] - mg(h - l\cos\theta)$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) - \frac{\partial L}{\partial \theta} = 0$$

$$\frac{\partial L}{\partial \dot{\theta}} = m l \dot{\theta} \rightarrow \frac{d}{dt} = m l \ddot{\theta}$$

$$\frac{\partial L}{\partial \theta} = m l \left(\Omega^2 (w + l\sin\theta)\cos\theta - g\sin\theta \right)$$

$$= m l \ddot{\theta} - m l \left[\Omega^2 (w + l\sin\theta)\cos\theta - g\sin\theta \right]$$

$$= l\ddot{\theta} + g\sin\theta - \Omega^2 (w + l\sin\theta)\cos\theta = 0$$

$$\rightarrow \boxed{\ddot{\theta} = \frac{\Omega^2}{l} (w + l\sin\theta)\cos\theta - \frac{g}{l}\sin\theta}$$

- positions of x y and z (y and z are switched in code)

$$x(t) = (w + l\sin(\theta(t)))\cos(\Omega t)$$

$$y(t) = -l\cos(\theta(t))$$

$$z(t) = (w + l\sin(\theta(t)))\sin(\Omega t)$$